# Problem 3.0.16: Evaluating the Gaussian Double Integral

## Question, Solution (All Methods), Python Code & Plots IN 2007

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# Question: The Problem Statement

**Evaluate the double integral:**

$$I = \int_0^\infty \int_0^\infty e^{-x^2-y^2}\, dx\, dy$$

**Given:**

- Region of integration: **First Quadrant** $(x \geq 0, y \geq 0)$
- Integrand: $e^{-x^2-y^2}$ (always positive and symmetric)
- Find the exact value of this integral

**Source:** IN 2007, Problem 3.0.16

# Solution - Method 0: Matrix Formulation

## Step 1: Matrix Representation

$$x^2 + y^2 = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \boldsymbol{x}^T A \boldsymbol{x}$$

where $A = I$ (Identity matrix), symmetric and positive definite.

## Step 2: Standard Gaussian Result

$$\int_{\mathbb{R}^n} e^{-\boldsymbol{x}^T A \boldsymbol{x}} d\boldsymbol{x} = \sqrt{\frac{\pi^n}{\det(A)}}$$

For $n = 2$, $A = I$, $\det(A) = 1$:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dx\, dy = \pi$$

# Solution - Method 0: Matrix Formulation (cont.)

**Step 3: Restricting to First Quadrant**

- Given region: $(x \geq 0, y \geq 0)$ is **one-fourth** of full plane
- Function is symmetric in all quadrants

$$I = \frac{1}{4} \times \pi = \boldsymbol{\frac{\pi}{4}}$$

**Result:** $\boxed{I = \frac{\pi}{4} \approx 0.7854}$

# Solution - Method 1: Polar Coordinates

**Key Idea:** Convert Cartesian to polar coordinates

**Transformation:**

$$x = r\cos\theta, \quad y = r\sin\theta, \quad dA = r\,dr\,d\theta$$

$$x^2 + y^2 = r^2$$

**Integral becomes:**

$$I = \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} r\,dr\,d\theta$$

**Step 1: Separate the integrals**

$$I = \left(\int_0^{\pi/2} d\theta\right)\left(\int_0^{\infty} re^{-r^2} dr\right)$$

# Solution - Method 1: Polar Coordinates (cont.)

**Step 2: Evaluate angular part**

$$\int_0^{\pi/2} d\theta = \frac{\pi}{2}$$

**Step 3: Evaluate radial part**

Substitution: $u = r^2$, $du = 2r\,dr$

$$\int_0^{\infty} re^{-r^2} dr = \frac{1}{2}\int_0^{\infty} e^{-u} du = \frac{1}{2}$$

**Result:**

$$I = \frac{\pi}{2} \times \frac{1}{2} = \boldsymbol{\frac{\pi}{4}}$$

# Solution - Method 2: Separation of Variables

**Key Idea:** Express as product of independent integrals

**Observation:**

$$e^{-x^2-y^2} = e^{-x^2} \cdot e^{-y^2}$$

**Factorization:**

$$I = \int_0^\infty \int_0^\infty e^{-x^2} e^{-y^2} \, dx \, dy$$

$$= \left( \int_0^\infty e^{-x^2} dx \right) \left( \int_0^\infty e^{-y^2} dy \right)$$

Since both integrals are identical:

$$I = \left( \int_0^\infty e^{-x^2} dx \right)^2$$

# Solution - Method 2: Separation of Variables (cont.)

**Standard Gaussian Integral:**

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$$

**Derivation:** Let $I_1 = \int_0^\infty e^{-x^2}\,dx$

$$I_1^2 = \left(\int_0^\infty e^{-x^2}\,dx\right)\left(\int_0^\infty e^{-y^2}\,dy\right) = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dx\,dy$$

Converting to polar: $I_1^2 = \frac{\pi}{4}$, thus $I_1 = \frac{\sqrt{\pi}}{2}$

**Final Result:**

$$I = \left(\frac{\sqrt{\pi}}{2}\right)^2 = \boldsymbol{\frac{\pi}{4}}$$

# Solution - Method 3: Error Function Approach

**Error Function Definition:**

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

**Key Property:**

$$\lim_{x\to\infty} \operatorname{erf}(x) = 1 \quad \Rightarrow \quad \int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$$

**Express integral using erf:**

$$I = \lim_{a,b\to\infty} \int_0^a \int_0^b e^{-x^2-y^2} dx\, dy$$

$$= \lim_{a,b\to\infty} \left( \int_0^a e^{-x^2} dx \right) \left( \int_0^b e^{-y^2} dy \right)$$

$$= \lim_{a,b\to\infty} \frac{\sqrt{\pi}}{2} \operatorname{erf}(a) \cdot \frac{\sqrt{\pi}}{2} \operatorname{erf}(b) = \boldsymbol{\frac{\pi}{4}}$$

# Solution - Method 4: Monte Carlo Numerical Integration

**Key Idea:** Use random sampling to approximate the integral

**Monte Carlo Method:**

- Generate random points $(x_i, y_i)$ in region $[0, L] \times [0, L]$
- Evaluate $f(x_i, y_i) = e^{-x_i^2 - y_i^2}$ at each point
- Average the function values and multiply by area

**Formula:**

$$I \approx L^2 \times \frac{1}{N} \sum_{i=1}^{N} e^{-x_i^2 - y_i^2}$$

For large $L$ and $N \to \infty$:

$$I \approx \frac{\pi}{4} \approx 0.7854$$

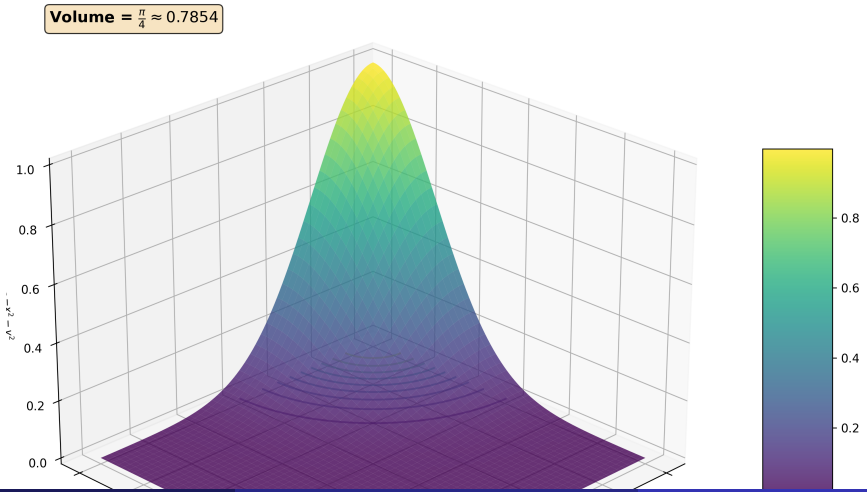
# Solution Summary: All Methods Comparison

| Method | Key Technique | Difficulty |
|---|---|---|
| Matrix Formulation | Gaussian integral theorem | Easy |
| Polar Coordinates | Change of variables | Medium |
| Separation of Variables | Product factorization | Easy |
| Error Function | Special function limits | Medium |
| Monte Carlo | Numerical simulation | Easy (coding) |

**All methods converge to:** $\boxed{\frac{\pi}{4}}$

# Visualization 1: 3D Surface Plot

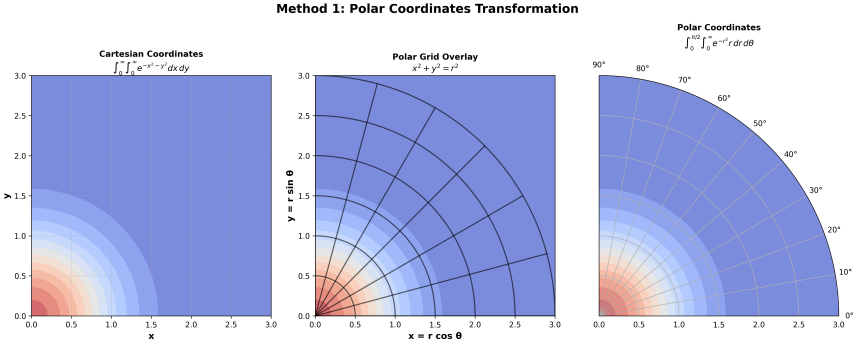
**Surface:** $z = e^{-x^2-y^2}$ **(First Quadrant)**



# Python Code: Surface Plot

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Create mesh grid
x = np.linspace(0, 3, 100)
y = np.linspace(0, 3, 100)
X, Y = np.meshgrid(x, y)

# Gaussian function
Z = np.exp(-X**2 - Y**2)

# Create 3D plot
fig = plt.figure(figsize=(12, 9))
ax = fig.add_subplot(111, projection='3d')
surf = ax.plot_surface(X, Y, Z, cmap='viridis',
                       alpha=0.8)
```

# Visualization 2: Polar Coordinates Transformation

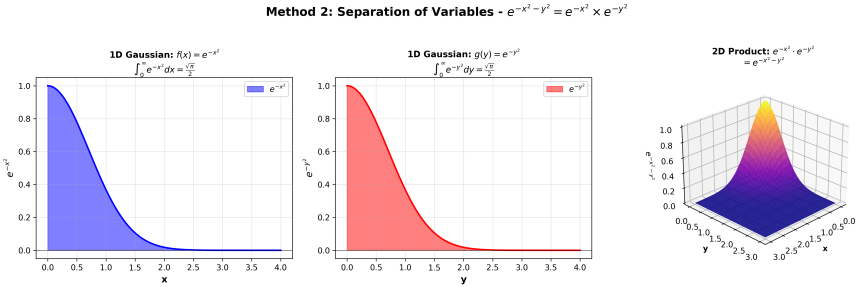



**Method 1:** Polar coordinate transformation in first quadrant

# Python Code: Polar Visualization

```python
import numpy as np
import matplotlib.pyplot as plt

# Polar grid
theta = np.linspace(0, np.pi/2, 50)
r = np.linspace(0, 3, 50)

# Create subplots showing transformation
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

# Plot 1: Cartesian grid
# Plot 2: Polar grid overlay
# Plot 3: Pure polar representation

for ax in axes:
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.3)
```

# Visualization 3: Separation of Variables

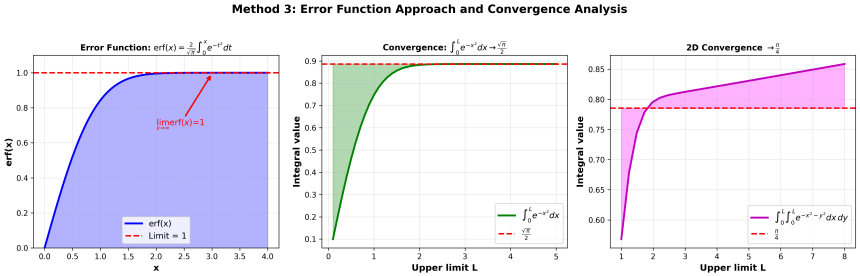



**Method 2:** Product of two 1D Gaussian functions

# Python Code: Separation of Variables

```python
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 3, 1000)
gaussian_1d = np.exp(-x**2)

# Create visualization
fig, axes = plt.subplots(2, 2, figsize=(12, 10))

# Plot 1D Gaussians and their product
axes[0, 0].plot(x, gaussian_1d, 'b-', linewidth=2)
axes[0, 0].set_title(r'$e^{-x^2}$')

axes[0, 1].plot(x, gaussian_1d, 'r-', linewidth=2)
axes[0, 1].set_title(r'$e^{-y^2}$')

# 2D product visualization
```

# Visualization 4: Error Function Convergence

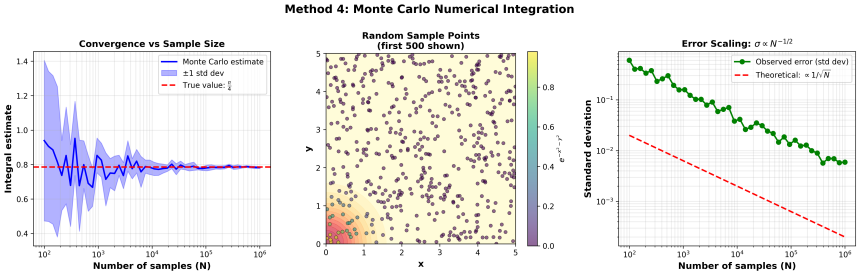



**Method 3:** Error function convergence to limit

# Python Code: Error Function

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import erf

x = np.linspace(0, 5, 1000)
erf_values = erf(x)

fig, axes = plt.subplots(2, 1, figsize=(12, 10))

# Plot error function
axes[0].plot(x, erf_values, 'b-', linewidth=2)
axes[0].axhline(y=1, color='r', linestyle='--',
                label=r'$\lim_{x\to\infty} erf(x) = 1$')
axes[0].set_title('Error Function')

# Plot integral approximation
# ...
```

# Visualization 5: Monte Carlo Convergence

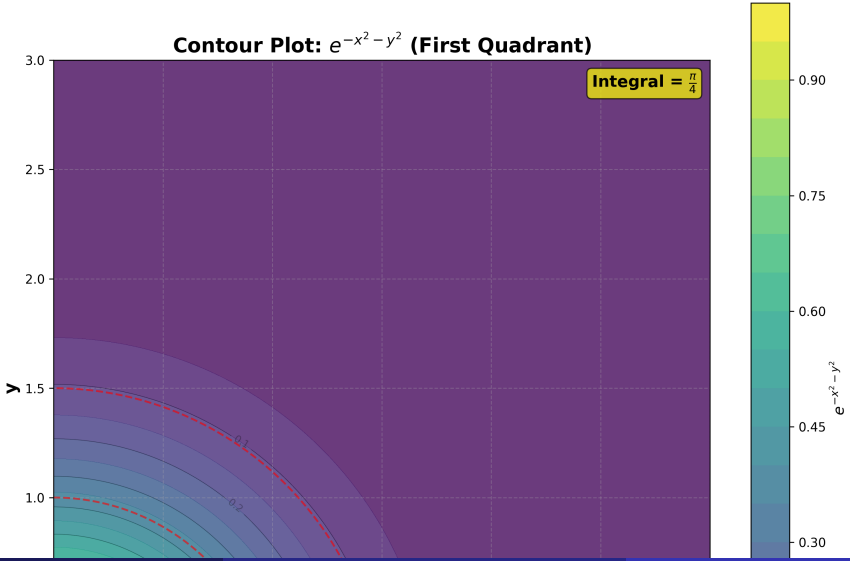



**Method 4:** Monte Carlo simulation converging to $\pi/4$

# Python Code: Monte Carlo Integration

```python
import numpy as np
import matplotlib.pyplot as plt

def monte_carlo_integral(N, L=10):
    x = np.random.uniform(0, L, N)
    y = np.random.uniform(0, L, N)
    f_values = np.exp(-x**2 - y**2)
    return L**2 * np.mean(f_values)

# Convergence analysis
sample_sizes = np.logspace(2, 6, 50, dtype=int)
estimates = [monte_carlo_integral(N) for N in sample_sizes]

plt.plot(sample_sizes, estimates, 'b-', linewidth=2)
plt.axhline(y=np.pi/4, color='r', linestyle='--',
            label=r'True value: $\pi/4$')
plt.xscale('log')
plt.savefig('./fig/monte_carlo_convergence.eps', dpi=300)
```

# Visualization 6: Contour Plot



# Python Code: Contour Plot

```python
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 3, 200)
y = np.linspace(0, 3, 200)
X, Y = np.meshgrid(x, y)
Z = np.exp(-X**2 - Y**2)

fig, ax = plt.subplots(figsize=(10, 8))
contour = ax.contour(X, Y, Z, levels=15, cmap='viridis')
contourf = ax.contourf(X, Y, Z, levels=15,
                       cmap='viridis', alpha=0.6)
plt.colorbar(contourf)
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title(r'Contour Plot: $e^{-x^2-y^2}$')
plt.savefig('../figs/contour_plot.png', dpi=300)
```

# Master Script: Generate All Plots

```python
"""
Master script to generate all visualizations
for Problem 3.0.16
"""
import subprocess
import sys

scripts = [
    'surface_plot.py',
    'contour_plot.py',
    'polar_visualization.py',
    'separation_variables.py',
    'error_function.py',
    'monte_carlo_convergence.py'
]

for script in scripts:
    print(f"Running {script}...")
```

# Final Answer

$$\boxed{\int_0^\infty \int_0^\infty e^{-x^2-y^2}\, dx\, dy = \frac{\pi}{4}}$$

- **Exact value:** $\pi/4$
- **Approximate value:** 0.785398...
- **Methods verified:** 5 (Matrix, Polar, Separation, Error Function, Monte Carlo)
- **Python visualizations:** 6 plots generated

# Conclusion

**Key Takeaways:**

- Used **matrix formulation** of quadratic form
- Applied **Gaussian integral result**
- Explored **4 analytical methods** + 1 numerical method
- Used **symmetry argument** for region restriction
- Verified with **Python visualizations**

**Final Result:** $\frac{\pi}{4}$

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